Intra-Cell Spatial Reuse in LEO Downlinks: Beam-Leakage-Aware Satellite Scheduling

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LEGACY • ITU EPFD / FCC 2022
1 CO-FREQUENCY BEAM
Starlink Gen2 license: 12.2–12.7 and 19.7–20.2 GHz.
GSO arc: ≥18° at full power; reduced power at smaller angles (EPFD).

09 JAN 2026 • FCC WAIVER
UP TO 8 OVERLAPPING BEAMS
SpaceX Ku-band operation in the U.S.
4° GSO-arc avoidance; interference safeguards.

30 APR 2026 • FCC 26-26 ADOPTED
NO FIXED BEAM-COUNT CAP
U.S. alternative based on GSO-link performance;
≥3° GSO-arc avoidance. Released 1 May 2026.

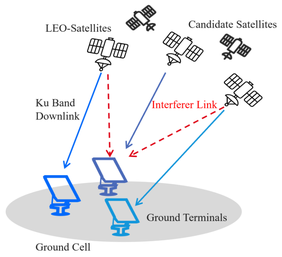
+27.2% Mean per-beam capacity
362.9 → 461.6 Mbps • 8 beams

2.31× Mean per-snapshot minimum rate
173.8 → 401.3 Mbps • 8 beams

267× Faster than exhaustive search
529.48 → 1.98 ms • select 8 of 28

1 SCENARIO & LINK MODEL

Shared hotspot, shared spectrum, cochannel interference.



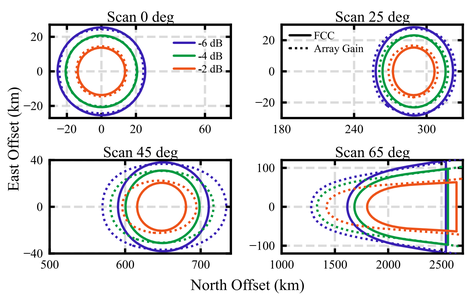
TRACE & RADIO SETTINGS

Trace	24 h / 720 snapshots
Visible	49–90; elevation ≥25°
f / BW	11.7 GHz / 250 MHz
TX / RX	30×30 / 20×23 URAs
Channel	LoS + ITU-R P.618
Noise	5-dB receiver NF

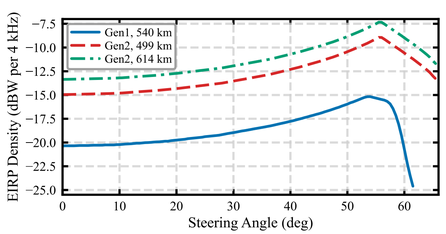
Independent streams; matched steering; 2-min samples.

Array and EIRP calibration

SpaceX FCC contours and local satellite 30 x 30 array footprint
11700 MHz | Altitude 614 km



Solid: filed contours; dotted: 30×30 array model.
Agreement is closest at 0°/25°; modeled footprints widen at 45°/65°.



Gen-1/Gen-2 EIRP; TX/RX gains: 34/30.4 dBi.

From ephemerides to capacity

Geometry → channel & power → BL-BnB selection
User-specific beams → SINR → capacity

Select at the cell reference; evaluate per-user rates.
Rank-one LoS steering ≡ SVD; P.618: 1% exceedance.

2 LEAKAGE-AWARE SCHEDULING

Select K satellites jointly by received power and aggregate leakage into each receive beam.

$$\eta_{s,j} = \frac{|\mathbf{a}_{r,s}^H \mathbf{a}_{r,j}|^2}{N_r^2} \qquad \gamma_s(\mathcal{A}) = \frac{q_s}{\sigma^2 + \sum_{j \in \mathcal{A}, j \neq s} q_j \eta_{s,j}}$$

$$\mathcal{A}^\star = \arg\max_{\mathcal{A} \subseteq \mathcal{S},\ |\mathcal{A}| = K} \sum_{s \in \mathcal{A}} r(\gamma_s(\mathcal{A}))$$

q_s: desired power; $\eta_{s,j}$: normalized RX-array leakage.
Study rate: $r(\gamma) = 0.6B \log_2(1 + \min\{\gamma, 10^3\})$.

BL-BnB: exact search with admissible pruning

Initialize Greedy + one-swap; test top-/bottom-K power sets.

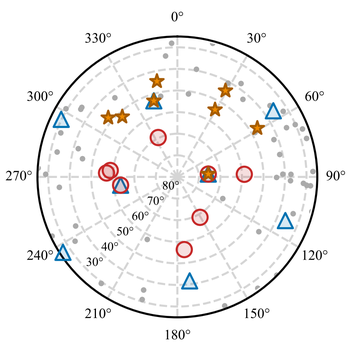
Bound Sum optimistic rates using the smallest possible future interference from the remaining candidates.

Search Include or exclude a candidate; prune if its upper bound cannot improve the incumbent.

$$m = K - |\mathcal{A}|, \quad L_s(\mathcal{R}, m) = \text{sum of the } m \text{ smallest } q_j \eta_{s,j}$$
$$\mathrm{UB}(\mathcal{A}, \mathcal{R}) = \sum_{s \in \mathcal{A}} \bar{R}_s + \sum_{c \in \mathrm{Top}_m(\mathcal{R})} \bar{R}_c$$

Optimal for the supplied reference-point model.
6,480 exact benchmark matches; worst case remains exponential.

Different selected sets in the same sky

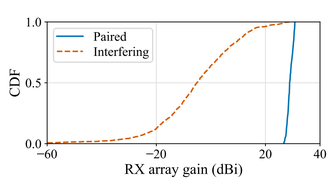


Nearest 8 favors high-elevation links.

BL-BnB trades link strength for lower aggregate leakage.

The two sets share only one satellite in this snapshot.

Snapshot 695: 71 visible satellites. Overlaid markers show shared choices.



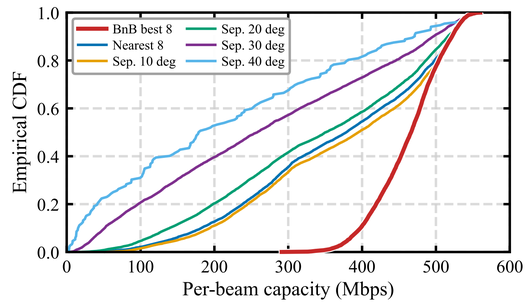
WHY LEAKAGE MATTERS

RX arrays resolve arrival directions, but residual leakage accumulates across active beams. Wider separation alone can sacrifice power.

Supporting archived receive-array sweep.

3 CAPACITY GAINS & COMPUTATIONAL COST

8 beams: balance power and interference

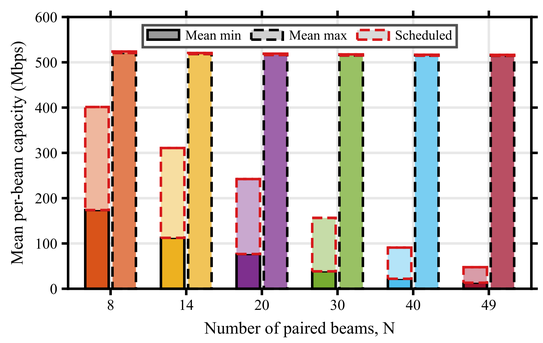


Mean per-beam capacity (Mbps)

Nearest	Sep. 10°	Sep. 20°	Sep. 30°	BL-BnB
362.9	372.9	339.8	267.7	**461.6**

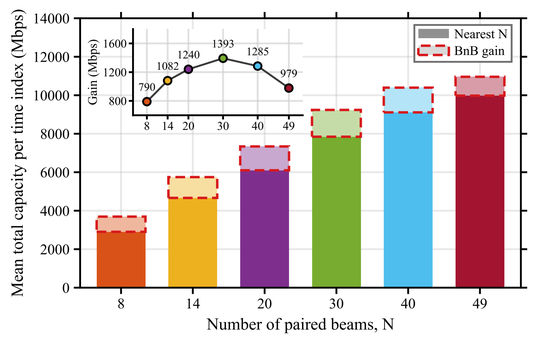
The 40° baseline has only 19/720 feasible snapshots.

Weaker links receive the largest gains



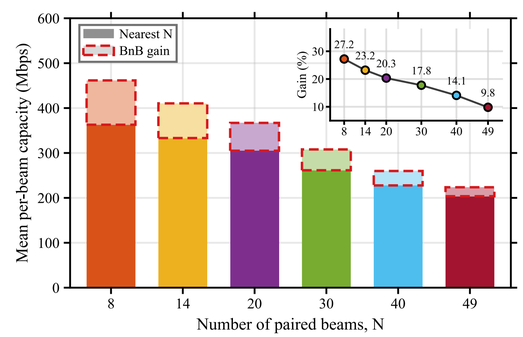
2.31× 8-beam mean min: 173.8 → 401.3 Mbps
Mean max: 520.6 → 523.8 Mbps

Total capacity grows sublinearly



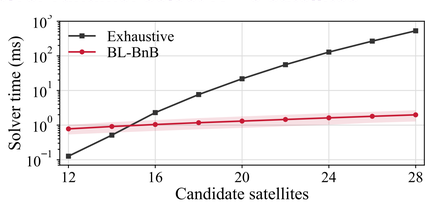
Nearest: 0.45 / 2.90 / 9.98 Gbps at N = 1 / 8 / 49.
BL-BnB absolute gain peaks at 1.393 Gbps for N = 30.

Per-beam gains diminish with N



+27.2% at 8 beams → +9.8% at 49 beams;
mean per-beam gain falls from 98.7 to 20.0 Mbps.

Solver runtime: select K = 8 satellites



Curves: 720 snapshots × 3 calls; mean ±1 SD.

FULL VISIBLE POOL · 49–90 CANDIDATES

4.83 ms
Median runtime

10.93 ms
95th-percentile runtime

PRACTICAL GATEWAY INPUTS
Ephemerides + terminal GPS
Beam powers + array responses
Precomputed P.618 loss tables
No inter-satellite carrier-phase information required.

Solver only; 3-call medians · 22 median nodes · 720/720 archived optima matched.

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The results are based on simulations and do not necessarily reflect the views of CableLabs.